Math Exercises

NMRC 2026

*Note: The exercises denoted with * are slightly more challenging.*

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1)

Plot the following groups of functions in the same graph:

a) $$y = x; \quad y = 2x; \quad y = 4x \tag{1}$$

b) $$y = \sqrt{x}; \quad y = x; \quad y = x^2 \tag{2}$$

c) $$y = e^{\frac{1}{2}x}; \quad y = e^{x}; \quad y = e^{2x} \tag{3}$$

2)

Compute the following derivatives:

$$\frac{d}{dx} 3x^x \tag{4}$$

$$f'(z), \text{ where } f(z) = 2e^z + \frac{1}{z^2} \tag{5}$$

$$\frac{d}{d\varphi} e^{4\varphi^2} \tag{6}$$

* $$\frac{d}{dx} \left(\cos^2(x) \right) \tag{7}$$

* $$\frac{d}{dx} \left(\frac{x}{1+e^{x^2}} \right) \tag{8}$$

3)*

Let $s(x) = \frac{1}{1+e^{-x}}$, i.e. the sigmoid function. Compute the derivative and show that $s'(x) = s(x)(1 - s(x))$ (note: this requires a bit of rewriting — there are various ways to get there, one of which involves cleverly adding $+1 - 1$). Finally, draw $s(x)$.

4)*

Consider a neuron modelled with a logistic function $f(x) = \frac{1}{1+e^{-kx}}$, with $k = 2$. Suppose you want this neuron to discriminate more strongly between negative and positive inputs: how would you change k to obtain this behaviour?

5)*

For the hyperbolic tangent function $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ compute the derivative and show that $\frac{d}{dx}\tanh(x) = 1 - \tanh^2(x)$.

6)

For the following functions, compute $\frac{\partial}{\partial x} f(x, y)$, $\frac{\partial}{\partial y} f(x, y)$, and (when applicable) $\frac{\partial}{\partial z} f(x, y)$:

$$f(x, y) = x^3 + 3xy + 4y^2 + x^2 y \tag{9}$$

$$f(x, y, z) = x^2 + 3xyz \tag{10}$$

$$f(x, y) = x^y \tag{11}$$

7)

Compute the gradient of the following functions:

$$f(x, y) = x^2 y \tag{13}$$

$$f(x, y, z) = x^2 y \cos z \tag{14}$$

$$f(x, y, z, t) = x^2 y t^{-1} \cos z \tag{15}$$

8)

Given the two following vectors $\mathbf{v}$ and $\mathbf{u}$, compute their dot product $\mathbf{v} \cdot \mathbf{u}$:

$$\mathbf{v} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \tag{17}$$

9)

Consider the two matrices A and B:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix} \tag{18}$$

Compute the following:

$$A + B \tag{19}$$

$$A + 2B \tag{20}$$

$$A^T - 2B \tag{21}$$